VACUTECH · PHASE 2 · BUSINESS INTELLIGENCE & DATA MINING

From RMAs to Recommendations

Three CRM questions, three transparent data-mining models

Course module: DLMDSEBA02 • Composite presentation

W1 · Apriori

24 rules from 2974 PM cases

Pareto: 20 SKUs cover ≈80%

W2 · Decision Tree

R^2 = 0.93 on 2000 test cases

MAE ≈ 1.07 days at intake

W3 · DT classifier

Threshold = 0.3, recall 49%

Accuracy 85% on 2000 test cases

All KPIs are read live from exports/metrics_summary.json

CRM tasks and the data-mining toolbox

Customer Relationship Management aims to identify, attract, retain and develop customers. Data mining supports each of these CRM tasks with a different family of algorithms[1)][14]. VacuTech's three W questions sit on the retention and development side: stock the right parts, make credible promises, and prevent rework before customers feel it.

Customer Identification

Segmentation, target selection

DM family: Clustering, classification

Customer Attraction

Direct marketing, lead scoring

DM family: Classification, association rules

Customer Retention

Service quality, churn, complaint triage

DM family: Classification (DT), regression

Customer Development

Cross-/up-sell, lifetime value, bundling

DM family: Association rules, regression

VacuTech mapping • W1 → Customer Development (parts bundling) • W2 → Customer Retention (credible ETAs) • W3 → Customer Retention (QA-failure prevention)

[14] Ngai et al. 2009 [1] Shearer 2000

Which add-on parts should be stocked or bundled for PM visits?

Business question

During preventive-maintenance jobs, technicians often consume parts that are NOT in the standard PM kit — forcing a second trip and eroding customer trust. We want a short, ranked stocking shortlist and a set of bundle rules a planner can act on.

Data used

- 538 PM cases from repairs.csv (maintenance_kit_applied = 1)
- Add-on lines from parts_used.csv (kit_part_flag = 0)
- Pivoted into a boolean basket matrix: 538 cases x 36 add-on parts
- No imputation, no scaling — Apriori works on boolean baskets

Apriori association rules + Pareto

- Apriori (mlxtend) discovers frequent co-occurring add-on parts
- Rules ranked by lift, retained when lift exceeds independence
- Pareto chart on part counts complements rules with a SKU shortlist

Fixed hyperparameters

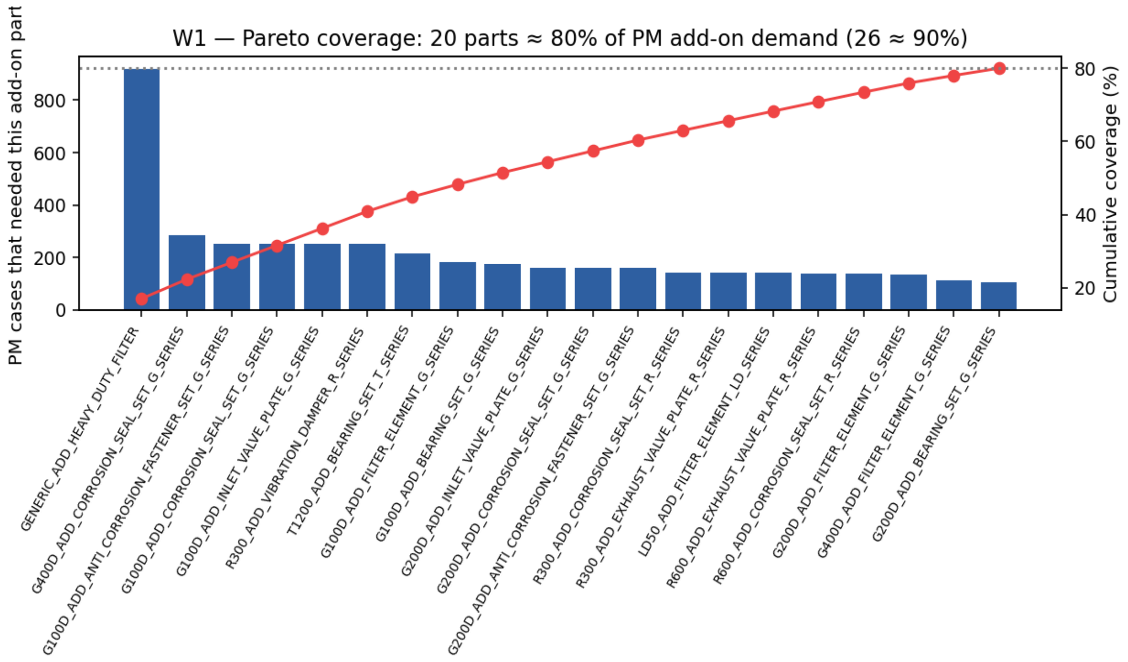
- min_support = 0.05 (≈ 25-50 baskets at this dataset size)
- min_lift = 1.2 (meaningfully stronger than chance)
- Outputs: w1_rules.csv, w1_metrics.json, two PNG charts

Frameworks

mlxtend.apriori · mlxtend.association_rules

[3] Agrawal & Srikant 1994 [4] Brin et al. 1997

Bundle rules + a 21-SKU Pareto shortlist



Rules found	Top lift	Top confidence	Pareto 80% / 90%
24	18.59	100%	20 / 26

Management action

- Lock the Pareto-head SKUs at every regional warehouse
- Pilot one bundle rule per pump series next to the standard PM kit
- Track the 'first-visit completion' rate before vs after roll-out

[6] Wang et al. 2018 [7] Didriksen et al. 2026

At intake, how long will this repair take?

Business question

The service planner must commit to a customer-facing window. The model must use only intake-safe features (no QA or post-repair fields) and remain explainable to a front-desk advisor.

Data used

- Target: repair_duration_days (right-skewed)
- Numeric: pump_age_years, technician_experience_years, parts_cost_eur
- Categorical: pump_model, complexity_class, failure_type, parts_from_hq_flag, region
- Preparation: dropna + one-hot encoding (no scaler)

Decision Tree Regressor (CART)

- One interpretable algorithm — splits readable on a whiteboard
- Trees are scale-free and need no target transform
- 80/20 train/test split, seed = 7, evaluated on held-out data

Fixed hyperparameters

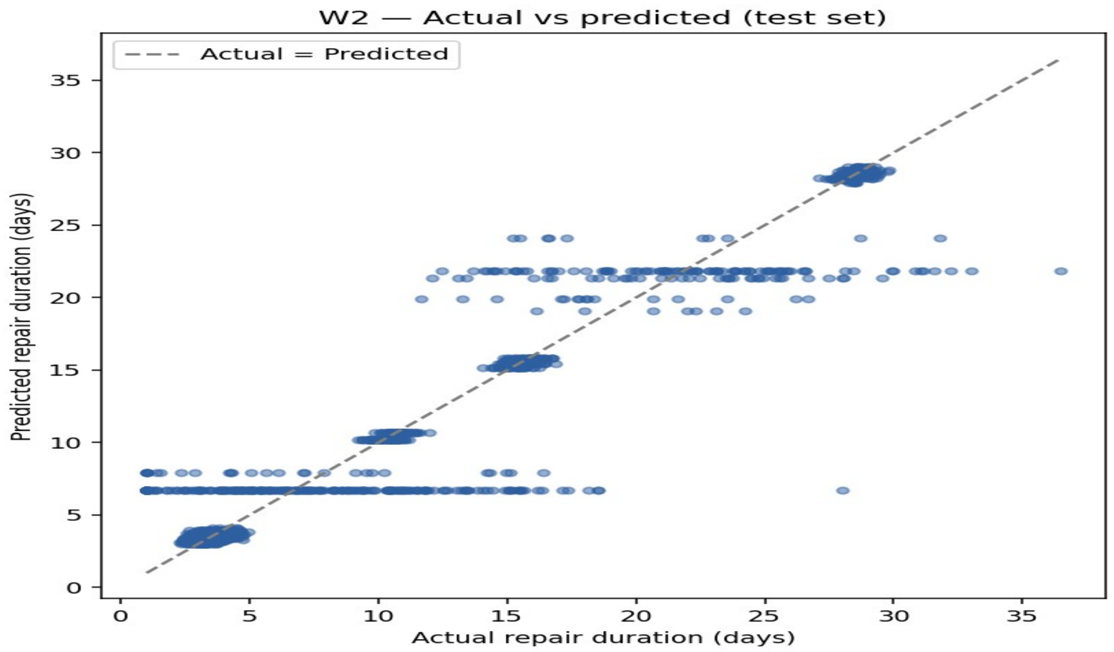
- max_depth = 6 (captures pump x complexity interactions)
- min_samples_leaf = 20 (≥ 20 cases per leaf)
- random_state = 7 (deterministic split)

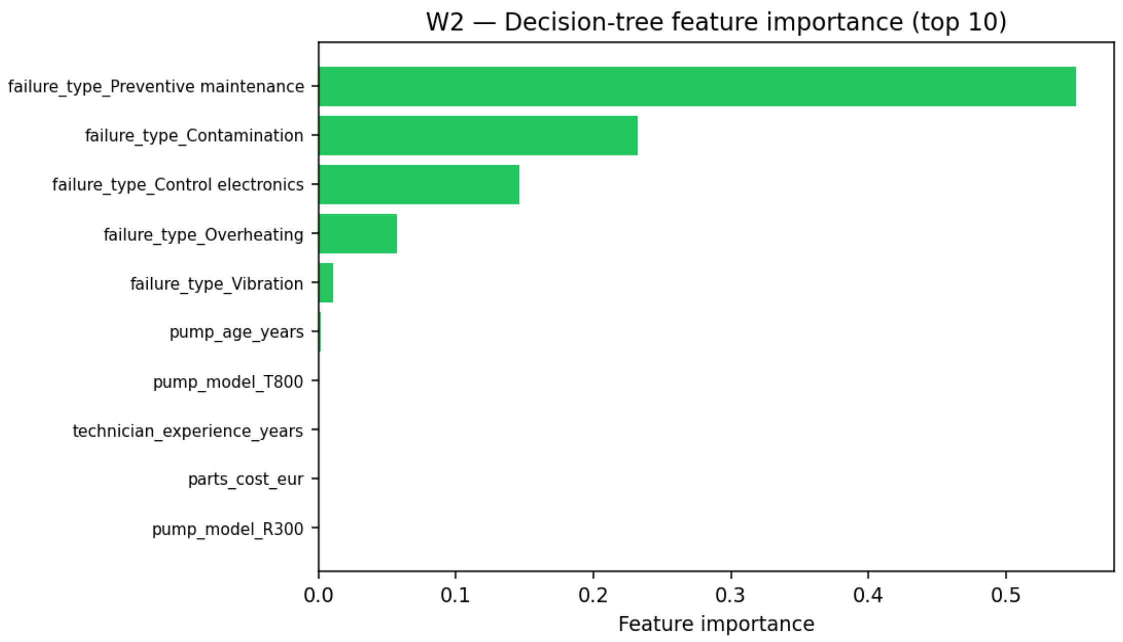
Frameworks

scikit-learn DecisionTreeRegressor · CART (Breiman et al. 1984)

[8] Breiman et al. 1984 [9] Rudin 2019

Usable intake-time ETAs — publish as a +/- range





Train / Test	R^2	MAE (days)	RMSE (days)
8000 / 2000	0.927	1.07	2.20

Management action

- Publish predicted duration as a ±2-day range on the customer acknowledgement
- Use feature importances as a routing signal (long jobs → senior techs)
- Refresh the model quarterly; track MAE drift

[10] Jang et al. 2021 [11] Ding et al. 2022 · Acela

Which finished repairs deserve a closer pre-QA look?

Business question

Before QA sign-off, operations wants a single yes/no flag per job. The target is imbalanced (~15% failures), so we use a deliberate operating threshold instead of bare accuracy.

Data used

- Target: qa_failed_flag (binary; ≈15% positives)
- Numeric: pump_age_years, technician_experience_years, parts_cost_eur, repair_duration_days
- Categorical: pump_model, complexity_class, failure_type, parts_from_hq_flag
- Stratified 80/20 split on the failure flag

Decision Tree Classifier + fixed threshold

- predict_proba -> single operating threshold = 0.30
- One algorithm, one operating point = governable dashboard flag
- Evaluation: confusion matrix, recall + precision (not accuracy alone)

Fixed hyperparameters

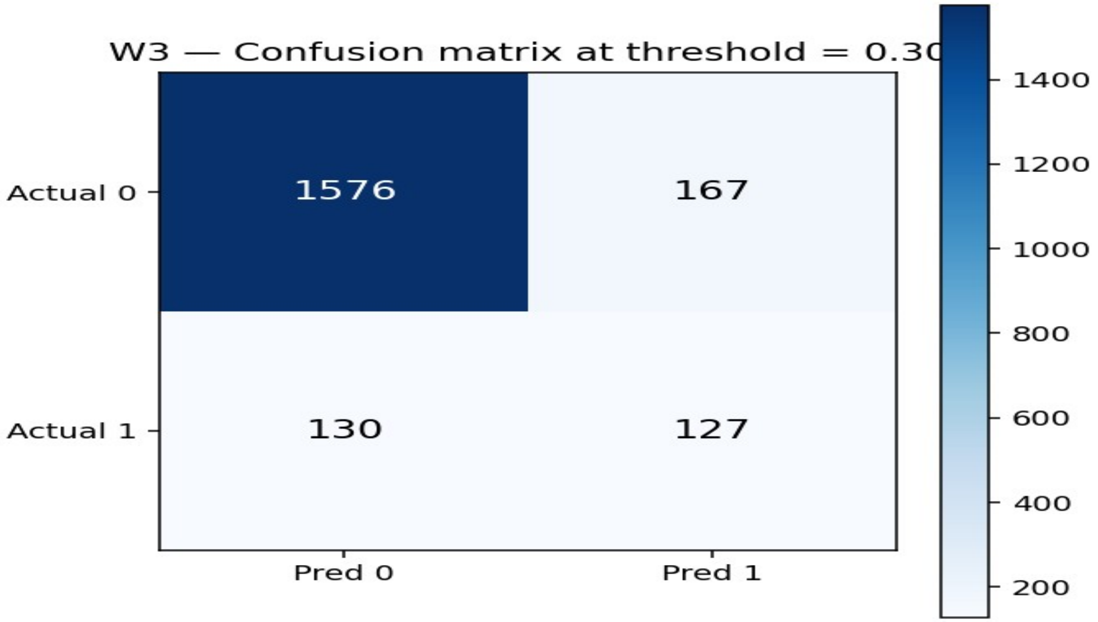
- max_depth = 6 ; min_samples_leaf = 20 ; random_state = 7
- threshold = 0.30 (balances recall vs review capacity)
- stratify = y to preserve class balance in train/test

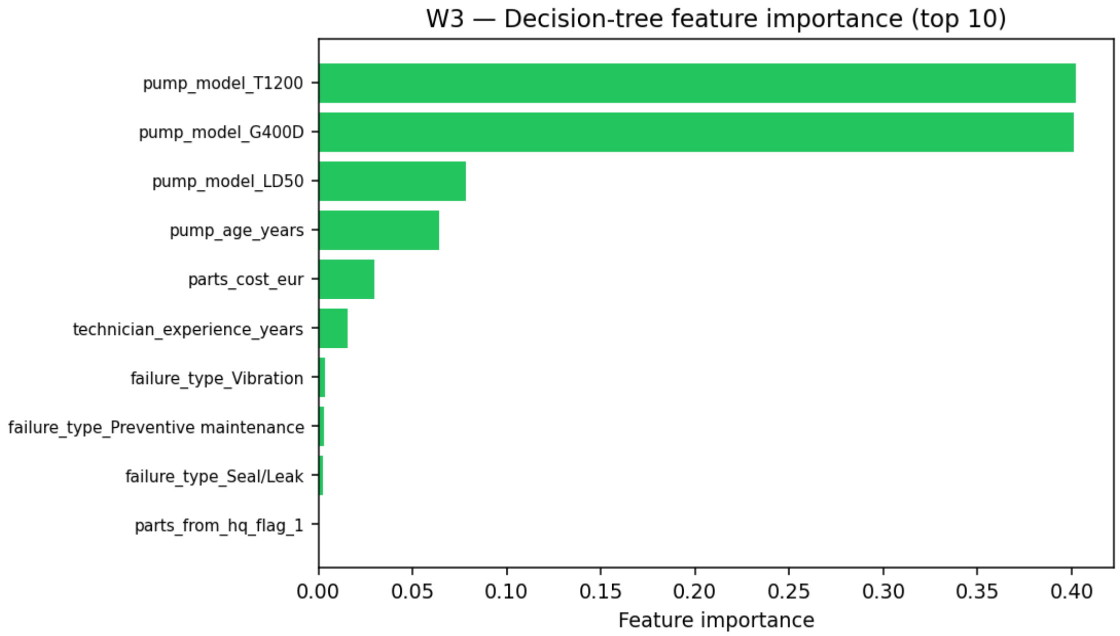
Frameworks

scikit-learn DecisionTreeClassifier · Imbalanced classification metrics

[8] Breiman et al. 1984 [9] Rudin 2019

A governable pre-QA triage flag at threshold 0.30





Threshold	Recall (failure)	Precision (failure)	Accuracy
0.3	49%	43%	85%

Management action

- Route flagged jobs through a 5-minute senior-tech pre-QA review
- Monitor the flag-vs-failure ratio weekly and adjust the threshold
- Avoid post-repair-only features when scoring at intake-time triage

[12] Zhang et al. 2023 [13] Sonawani & Mukhopadhyay 2013

CONCLUSION

Three concrete recommendations for VacuTech management

W1 · Inventory

Pre-stock the top 20 add-on SKUs at every regional warehouse and pilot one bundle rule per pump series. Apriori found 24 rules with top lift 18.6.

CRM lens: customer development — first-visit completion + cross-sell of bundled BOMs.

W2 · Repair ETA

Publish duration on the customer acknowledgement as a ±2-day range. The shallow tree reaches $R^2 \approx 0.93$ and MAE ≈ 1.07 days, while staying readable to the front-desk advisor.

CRM lens: customer retention — credible promises, fewer status calls.

W3 · QA triage

Send the jobs flagged at threshold 0.3 through a 5-minute senior pre-QA review. Recall ≈ 49% at 85% overall accuracy makes the workload manageable.

CRM lens: customer retention — prevent repeat-RMA situations.

Why interpretable models? Stakeholders (planners, advisors, QA supervisors) must act on the output. A shallow, auditable model that they can read and override is preferable to a high-accuracy black box they cannot challenge[9-2].

[9] Rudin 2019 [2] CRISP-DM Consortium 2000

Peer-reviewed sources and frameworks

[1] Shearer, C. (2000). The CRISP-DM model: The new blueprint for data mining. Journal of Data Warehousing, 5(4), 13-22. link

[2] CRISP-DM Consortium (2000). CRISP-DM 1.0: Step-by-step data mining guide. link

[3] Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules in large databases. Proc. VLDB '94, 487-499. link

[4] Brin, S., Motwani, R., Ullman, J. D., & Tsur, S. (1997). Dynamic itemset counting and implication rules for market basket data. Proc. ACM SIGMOD, 255-276. link

[5] Mundler, N. (2019). Association rule mining and itemset-correlation based variants. arXiv:1907.09535. link

[6] Wang, J., Pan, X., Wang, L., & Wei, W. (2018). Method of spare parts prediction models evaluation based on grey comprehensive correlation degree and association rules mining: A case study in aviation. Mathematical Problems in Engineering, 2018, 2643405. link

[7] Didriksen, S. K., Sigsgaard, K. W., Mortensen, N. H., & Jespersen, C. B. (2026). Assigning spare parts management decision-making strategies: A holistic portfolio classification methodology. Applied Sciences, 16(4), 1961. link

[8] Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). Classification and Regression Trees. Chapman & Hall. link

[9] Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 1, 206-215. link

[10] Jang, J., Nana, D., Hochschild, J., & de Lorenzo, J. V. H. (2021). Predicting breakdown risk based on historical maintenance data for Air Force ground vehicles. arXiv:2112.13922. link

[11] Ding, Y., Gao, A., Ryden, T., et al. (2022). Acela: Predictable datacenter-level maintenance job scheduling. arXiv:2212.05155. link

[12] Zhang, Y., Gao, Z., Sun, J., & Liu, L. (2023). Machine-learning algorithms for process condition data-based inclusion prediction in continuous-casting process. Sensors, 23(15), 6719. link

[13] Sonawani, S., & Mukhopadhyay, D. (2013). A decision tree approach to classify web services using quality parameters. arXiv:1311.6240. link

[14] Ngai, E. W. T., Xiu, L., & Chau, D. C. K. (2009). Application of data mining techniques in customer relationship management: A literature review and classification. Expert Systems with Applications, 36(2), 2592-2602. link

Frameworks used

- scikit-learn (Decision Trees)
- mlxtend (Apriori, association rules)
- pandas (data manipulation)
- Jupyter (CRISP-DM notebooks)
- CRISP-DM 1.0 guide

See ACADEMIC_METHOD_JUSTIFICATION.md for full per-W argumentation.